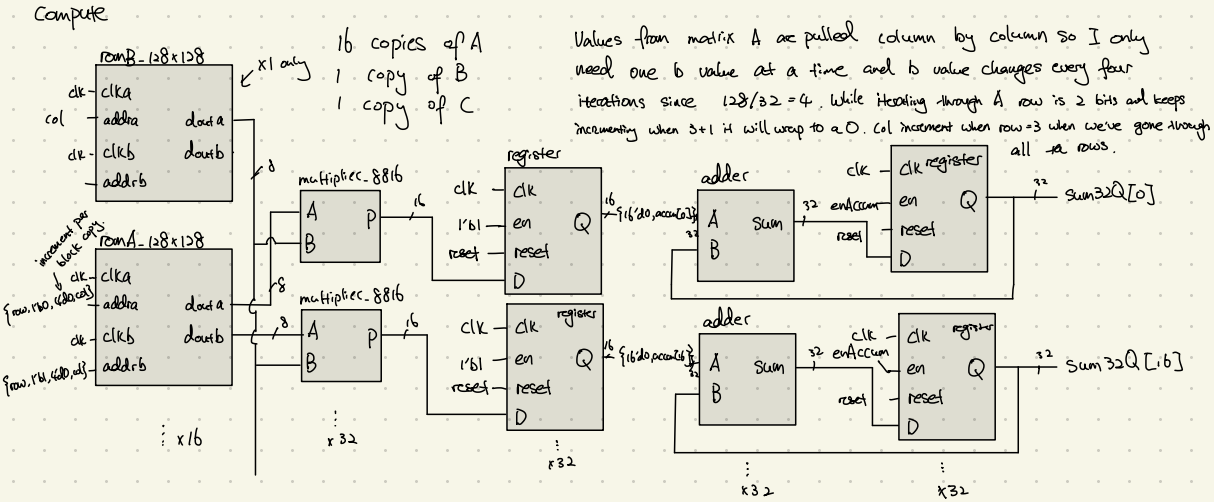
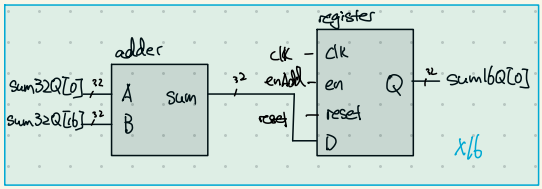
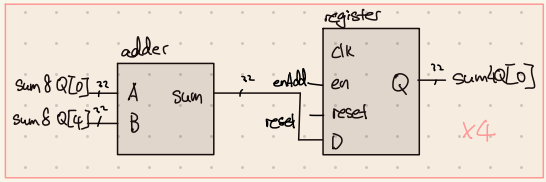
Compute

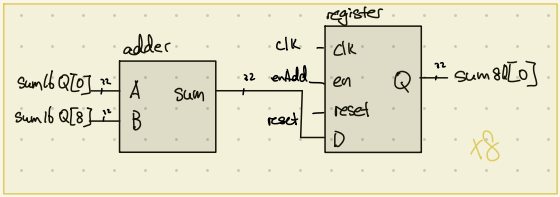
16 copies of A
1 copy of B
1 copy of C

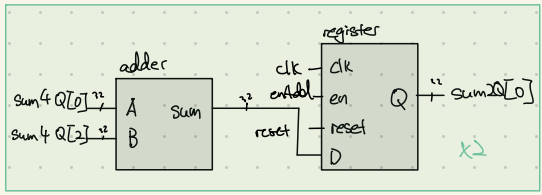
Values from matrix A are pulled column by column so I only need one b value at a time and b value changes every four iterations since 128/32 = 4. While iterating through A row is 2 bits and keeps incrementing when 3+1 it will wrap to a 0. Col increment when row=3 when we've gone through all the rows.

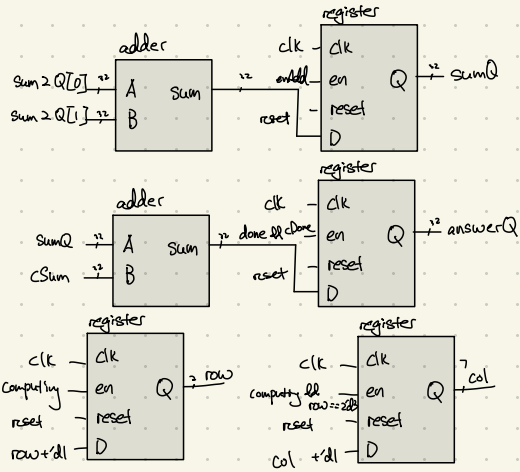




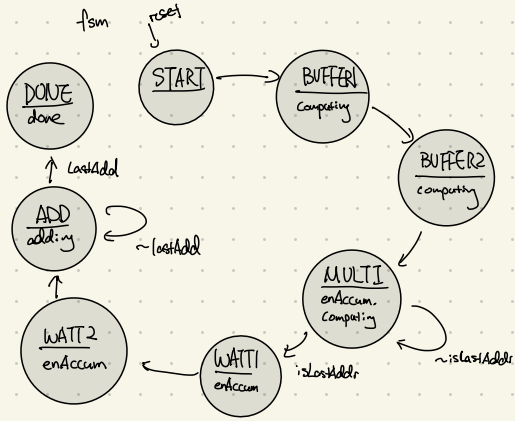






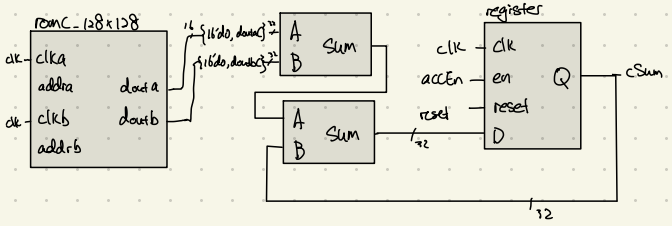


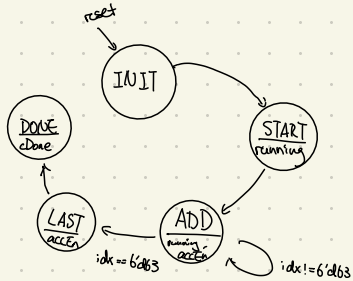
- This module outputs answerQ which is the sum to be displayed and clockCount which increments until and including the cycle that done is first asserted.
- enAdd is set to adding I kept both names since it feels more intuitive in the different contexts.

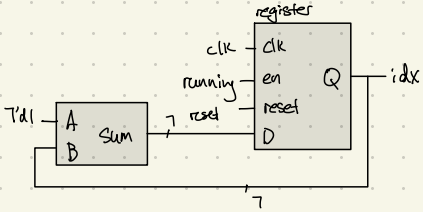


- isLastAdd is asserted when row == 2'd3 because the actual row it represent is 96-127 since 3·32 = 96 then we are pulling 32 values at a time.
- lastAdd is set when a counter that starts and continues counting while adding == 1 is equal to 4 because the counter is still 0 on the first cycle when adding is 1.

SumC


romC - 128x128
clka
addra
clkb
addrb
douta
doutb
16'd0, douta
16'd0, doutb
A
B
Sum
clk
accEn
reset
en
Q
D
cSum
register
32


reset
INIT
START
running
ADD
running
accEn
LAST
accEn
DONE
cDone
idx == 6'd63
idx != 6'd63


register
clk
en
reset
D
Q
idx
running
7'd1
A
B
Sum
7